MARVEL CINEMATIC UNIVERSE

Intelligence Dashboard & Success Prediction System

Comprehensive Data Analytics Project Report

49	5	3	6
MCU Titles	Phases	ML Models	Dashboard Pages

Tool Stack	Python · SQL · Scikit-learn · Power BI · HTML/JS
Dataset	49 MCU titles, Phase 1–5 (2008–2025)
Domain	Entertainment Analytics / Data Science
Report	8 Chapters · 30+ Pages

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Chapter 1

Introduction

1.1 Project Overview

This project presents a comprehensive data analytics study on the Marvel Cinematic Universe (MCU), encompassing 49 titles — movies and Disney+ series — across 5 phases from 2008 to 2025. The study applies the complete data analytics lifecycle: data collection, cleaning, exploratory analysis, SQL integration, machine learning, and interactive dashboard development, to extract meaningful business intelligence from MCU performance data.

The project culminates in a dual-dashboard system: a Power BI executive dashboard for business stakeholders, and a fully interactive web dashboard powered by a custom JARVIS AI voice assistant — combining technical depth with a Marvel-themed user experience.

1.2 Problem Statement

Despite MCU's unprecedented commercial success — generating over $31.96 billion in worldwide box office revenue — there is no unified analytical framework to evaluate movie performance across financial, critical, and audience dimensions simultaneously. Individual metrics such as box office revenue or IMDb ratings exist in isolation, failing to provide a holistic view of what makes an MCU title truly successful.

Key Question: What combination of factors — budget, phase, director, ratings — most accurately predicts the commercial and critical success of an MCU title?

1.3 Project Objectives

- Collect, clean, and prepare a comprehensive MCU dataset covering box office, ratings, budgets, and streaming engagement metrics.
- Perform in-depth exploratory data analysis (EDA) to uncover phase-wise trends, patterns, and correlations.
- Design a normalized SQL database with 5 relational tables and write advanced analytical queries including window functions, subqueries, and views.
- Engineer meaningful features including Profit, ROI, Success Category, and a custom Marvel Success Score KPI.
- Build machine learning models for revenue prediction (Random Forest Regressor), success classification (Random Forest Classifier), and unsupervised clustering (K-Means).
- Develop a Power BI executive dashboard with DAX-driven KPIs and multi-page reports.
- Build an interactive web dashboard with 6 pages, live filters, and a JARVIS AI voice assistant.
- Create a recommendation engine for MCU content discovery.

1.4 Scope of the Project

Included	Excluded
49 MCU titles (movies + Disney+ series)	Marvel comics and non-MCU films
Phase 1 through Phase 5 (2008–2025)	Sony Spider-Man and Fox X-Men films
Box office, IMDb, Rotten Tomatoes, budget, ROI data	Real-time Disney+ subscriber data
Python, SQL, Power BI, HTML/JS, Machine Learning	Deep learning and NLP models

1.5 Tools and Technologies

Tool / Library	Category	Purpose
Python (Pandas, NumPy)	Data Processing	Cleaning, EDA, feature engineering
Matplotlib, Seaborn	Visualization	Charts and heatmaps
MySQL	Database	Schema design and SQL queries
Scikit-learn	Machine Learning	Regression, classification, clustering
Power BI	Business Intel.	Executive dashboard and DAX KPIs
HTML / CSS / Chart.js	Web Development	Interactive web dashboard
Web Speech API	AI Assistant	JARVIS voice interface

1.6 Methodology

This project follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology. Starting from raw data collection, each phase builds upon the previous — cleaned data feeds into EDA, EDA insights guide SQL schema design, SQL outputs train machine learning models, and all results are visualized through dual dashboards. This structured approach ensures reproducibility and analytical rigour throughout the project.

Phase	Activity	Output
1	Data Collection & Cleaning	Clean MCU dataset (49 titles)
2	Python EDA	Statistical insights + visualizations
3	Feature Engineering	Profit, ROI, Success Score, categories

Phase	Activity	Output
4	SQL Database Design	5 normalized tables + 6 views
5	Machine Learning	3 trained models + cluster segments
6	Dashboard Development	Power BI + Interactive web dashboard

Chapter 2

Dataset Overview

2.1 Data Collection

The MCU dataset was compiled from multiple publicly available sources including IMDb, Box Office Mojo, Rotten Tomatoes, and The Numbers. The final dataset covers 49 Marvel Cinematic Universe titles released between 2008 and 2025, spanning 5 MCU phases and both theatrical movies and Disney+ streaming series.

49	5	33	16	17
Total Titles	MCU Phases	Movies	Series	Years Covered

2.2 Dataset Columns

Column	Type	Description
title	String	MCU title name
release_year	Integer	Year of release
mcu_phase_number	Integer	MCU phase (1–5)
imdb_rating	Float	IMDb user rating out of 10
rt_tomato_meter	Integer	Rotten Tomatoes critic score (%)
rt_audience_score	Integer	Rotten Tomatoes audience score (%)
production_budget	BigInt	Production budget in USD
worldwide_box_office	BigInt	Total worldwide box office in USD
domestic_box_office	BigInt	North America box office in USD
international_box_office	BigInt	International box office in USD
opening_weekend	BigInt	Opening weekend gross in USD
trailer_views	BigInt	Total YouTube trailer views
trailer_likes	BigInt	Total YouTube trailer likes
profit	BigInt	Profit = Box Office - Budget (engineered)
roi	Float	Return on Investment % (engineered)

Column	Type	Description
success_category	String	Blockbuster / Hit / Flop (engineered)
success_score	Float	Custom Marvel Success Score (engineered)
cluster	Integer	K-Means cluster label (0, 1, 2)
cluster_name	String	MCU Legends / Successful Heroes / Struggling Titles

2.3 Data Cleaning Steps

- Handled missing values using mean imputation for numerical columns (budget, ratings) and mode for categorical columns.
- Removed duplicate records using Pandas drop_duplicates() — no duplicates were found in the final dataset.
- Corrected data types: release_year converted to integer, monetary columns to int64.
- Standardized title formatting using str.title() for consistency.
- Validated revenue logic: all domestic + international values cross-checked against worldwide totals.
- Filtered out streaming-only series from box office analysis to avoid zero-revenue bias in regression models.

2.4 Feature Engineering

Four engineered features were created to enable deeper analysis and machine learning:

Profit

`Formula: worldwide_box_office - production_budget`

Net financial gain per title

ROI (%)

`Formula: (profit / production_budget) × 100`

Return on investment percentage

Success Category

`Formula: ROI-based labeling`

Blockbuster (ROI>500%) / Hit (ROI>100%) / Flop (ROI<0%)

Marvel Success Score

`Formula: IMDb×10 + Audience Score + Scaled ROI`

Custom composite KPI (0–100)

Chapter 3

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted using Python with Pandas, NumPy, Matplotlib, and Seaborn. The analysis covers revenue trends, rating distributions, phase performance, director comparisons, and feature correlations across all 49 MCU titles.

3.1 Phase-wise Performance Summary

Phase	Titles	Total Revenue	Avg IMDb	Avg ROI
1	6	$3.81 Billion	7.07	297%
2	6	$5.27 Billion	7.22	356%
3	11	$13.51 Billion	7.59	469%
4	15	$5.71 Billion	6.93	—
5	11	$3.66 Billion	6.40	—

Key Insight: Phase 3 is the MCU golden era — highest total revenue ($13.51B), best avg IMDb (7.59), and best avg ROI (469%). Phase 5 shows a clear quality and financial decline.

3.2 Top 10 Movies by Success Score

Rank	Title	Success Score	IMDb	Category
1	Spider-Man: No Way Home	97.95	8.2	Blockbuster
2	Avengers: Endgame	90.43	8.4	Blockbuster
3	Avengers: Infinity War	87.39	8.4	Blockbuster
4	The Avengers	83.54	8.0	Blockbuster
5	Spider-Man: Far From Home	79.78	7.4	Hit
6	Deadpool & Wolverine	79.05	7.5	Blockbuster
7	Guardians of the Galaxy	76.00	8.0	Hit
8	Captain America: Civil War	72.81	7.8	Hit
9	Thor: Ragnarok	73.37	7.9	Hit
10	Iron Man	73.29	7.9	Hit

3.3 Key Statistical Insights

Total worldwide box office	$31.96 Billion
Highest single title revenue	$2.80B — Avengers: Endgame
Highest ROI	860.9% — Spider-Man: No Way Home
Average IMDb rating	7.01 across all 49 titles
Blockbusters	8 titles (16.3% of total)
Flops	10 titles (20.4% of total)
Avg production budget	$191 Million
Avg worldwide revenue (movies only)	$878 Million

3.4 Correlation Analysis

A Pearson correlation analysis was conducted across numerical features to understand relationships between variables. Key findings from the correlation matrix:

- Production budget and worldwide box office show strong positive correlation ($r \approx 0.71$), confirming that higher-budget films tend to earn more.
- IMDb rating and RT audience score are strongly correlated ($r \approx 0.65$), indicating audience and user ratings align closely.
- Trailer views show moderate correlation with opening weekend revenue ($r \approx 0.55$), making it a useful predictor variable.
- Critics score (RT Tomato Meter) shows weaker correlation with revenue than audience score, suggesting commercial success is more audience-driven.
- ROI and production budget show slight negative correlation — smaller budget films can achieve higher percentage returns.

Chapter 4

SQL Analysis

A normalized relational database was designed in MySQL to store and query MCU data. The schema follows third normal form (3NF), separating concerns across 5 dedicated tables connected via foreign key relationships.

4.1 Database Schema

Table	Primary Key	Key Columns	Rows
movies	movie_id	title, release_year, phase_number, director	49
ratings	rating_id	imdb_rating, rt_tomato_meter, rt_audience_score	49
box_office	boxoffice_id	production_budget, worldwide_box_office, roi	49
streaming_stats	stream_id	trailer_views, trailer_likes, trailer_count	49
movie_success_metrics	metric_id	profit, roi, success_category	49

4.2 Key SQL Queries

Top 10 Movies by Revenue

```
SELECT m.title, b.worldwide_box_office FROM movies m JOIN box_office b ON m.movie_id
= b.movie_id ORDER BY b.worldwide_box_office DESC LIMIT 10;
```

Window Function — Phase-wise Rank

```
SELECT title, phase_number, RANK() OVER( PARTITION BY phase_number ORDER BY
worldwide_box_office DESC ) AS phase_rank FROM movies m JOIN box_office b ON
m.movie_id = b.movie_id;
```

Above Average Revenue — Subquery

```
SELECT title, worldwide_box_office FROM movies m JOIN box_office b ON m.movie_id =
b.movie_id WHERE worldwide_box_office > (SELECT AVG(worldwide_box_office) FROM
box_office);
```

4.3 Views Created

View Name	Purpose
marvel_executive_dashboard	Joins all 5 tables — title, IMDb, revenue, ROI, category
marvel_success_score	Calculates composite success score per title

View Name	Purpose
director_performance	Aggregates revenue, avg IMDb, avg ROI by director
phase_performance	Phase-level revenue, rating, and project count
audience_critic_gap	RT audience vs critic score difference per title

Chapter 5

Machine Learning

Three machine learning models were developed using Python's Scikit-learn library — a revenue prediction regressor, a success classifier, and an unsupervised clustering model. The feature set includes production budget, phase, IMDb rating, RT scores, and trailer views.

5.1 Revenue Prediction — Random Forest Regressor

A Random Forest Regressor was trained to predict worldwide box office revenue.

```
features = ['production_budget', 'mcu_phase_number', 'imdb_rating',
'rt_tomato_meter', 'trailer_views'] X_train, X_test, y_train, y_test =
train_test_split(X, y, test_size=0.2) rf = RandomForestRegressor(n_estimators=100,
random_state=42) rf.fit(X_train, y_train)
```

Baseline comparison with Linear Regression confirmed that the ensemble model captures non-linear budget-revenue relationships more effectively.

5.2 Success Classifier — Random Forest Classifier

A Random Forest Classifier was trained to predict the success category (Blockbuster / Hit / Flop) of each MCU title. Label encoding was applied to the target variable using Scikit-learn's LabelEncoder.

Note: Classifier accuracy was 50% — attributable to the small dataset size (n=49). With a larger dataset, accuracy would improve significantly. The clustering model compensates by providing unsupervised segmentation without labeling constraints.

5.3 K-Means Clustering

Unsupervised K-Means clustering (k=3) was applied to segment MCU titles into meaningful performance groups. The optimal k was determined using the Elbow Method. Features were standardized using StandardScaler before clustering.

Cluster	Label	Titles	Characteristics
0	MCU Legends	10	High revenue, high IMDb, top ROI — Endgame, NWH, IW
1	Successful Heroes	10	Mid-tier performers, mixed reception
2	Struggling Titles	29	Lower revenue or streaming-only, below-avg scores

Chapter 6

Power BI Dashboard

A 4-page Power BI executive dashboard was developed to present MCU analytics in a business intelligence format. Custom DAX measures power the KPIs, and interactive slicers enable filtering by Phase, Year, and Success Category.

6.1 Dashboard Pages

Page	Title	Key Visuals
1	Executive Dashboard	Total revenue KPI, phase trend, success distribution donut
2	Revenue Analysis	Budget vs revenue scatter, domestic vs intl split, ROI bar
3	Director Intelligence	Director revenue ranking, IMDb vs ROI scatter, performance matrix
4	MCU Segmentation	Cluster visualization, phase-wise cluster stacked bar

6.2 DAX Measures

Total Revenue

```
Total Revenue = SUM(movies[worldwide_box_office])
```

Total Profit

```
Total Profit = SUM(movies[profit])
```

Average ROI

```
Avg ROI = AVERAGE(movies[roi])
```

Average IMDb

```
Avg IMDb = AVERAGE(movies[imdb_rating])
```

Success Score

```
Success Score = [Avg IMDb] * 10 + [Avg Audience] + [Scaled ROI]
```

Chapter 7

Recommendation Engine

A content-based recommendation engine was built to suggest MCU titles based on similarity across multiple features. Given a seed title, the engine returns the top N most similar MCU titles ranked by composite similarity score.

7.1 Approach

The engine computes cosine similarity across a normalized feature matrix:

- IMDb rating and RT audience score (audience preference alignment)
- Production budget and ROI (financial profile)
- MCU phase number (narrative era proximity)
- Marvel Success Score (overall performance similarity)
- K-Means cluster label (segment-based grouping)

7.2 Sample Output

Example: Titles most similar to "Avengers: Infinity War":

Rank	Recommended Title	Similarity Score	Cluster
1	Avengers: Endgame	0.98	MCU Legends
2	Captain America: Civil War	0.91	MCU Legends
3	Black Panther	0.88	MCU Legends
4	Spider-Man: No Way Home	0.86	MCU Legends
5	Thor: Ragnarok	0.82	Struggling Titles

Chapter 8

Conclusion

8.1 Key Findings

- Phase 3 is the MCU golden era — highest total revenue ($13.51B), best avg IMDb (7.59), and best avg ROI (469%).
- Spider-Man: No Way Home achieved the highest Marvel Success Score (97.95) and the best ROI (860.9%) in the entire MCU.
- Avengers: Endgame is the highest-grossing MCU title at $2.80B worldwide.
- The Russo Brothers (Anthony & Joe Russo) are the top-performing director team by total revenue ($6.97B across 4 films).
- Phase 4 and 5 show a measurable quality and financial decline — lower avg IMDb, lower ROI, more flops.
- K-Means clustering revealed that only 10 of 49 titles (20%) qualify as "MCU Legends" — the true elite.
- Trailer views are a moderately strong predictor of opening weekend performance ($r \approx 0.55$).
- Budget does not guarantee success — Captain Marvel ($152M budget, 642% ROI) outperformed many larger-budget films.

8.2 Limitations

- Small dataset (n=49) limits ML model generalizability — classifier accuracy capped at 50%.
- Streaming series have no box office revenue, making financial comparisons between movies and series incomplete.
- Disney+ viewership data was unavailable, limiting streaming performance analysis.
- External factors (pandemic impact on Phase 4, competition from other franchises) were not modeled.

8.3 Future Scope

- Integrate real-time Disney+ viewership data via API for complete streaming analytics.
- Expand dataset to include DC Extended Universe for cross-franchise comparison.
- Implement NLP-based sentiment analysis on audience reviews from IMDb and Letterboxd.
- Deploy the web dashboard as a live public-facing application using Flask or FastAPI backend.
- Build a deep learning revenue prediction model using historical MCU data as time-series input.

This project demonstrates mastery of the complete data analytics lifecycle — from raw data collection through machine learning, business intelligence, and interactive dashboard *delivery — applied to a real-world entertainment domain with measurable, interview-ready insights.*